

Rimjhim Saxena

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EDUCATION

Ph.D. Economics, University of Colorado Boulder	May 2025 (Expected)
M.A. Economics, University of Colorado Boulder	2021
M.A. Economics, Boston University	2017
M.B.A. Finance, Thapar Institute of Engineering & Technology	2014
B.E. Industrial Engineering, Thapar Institute of Engineering & Technology	2014

RESEARCH AND TEACHING INTERESTS

Environmental and Resource Economics, Development Economics, Labor Economics

WORKING PAPERS

Barriers within Borders: Structural Transformation and Climate Change in India (Job Market Paper)
Stumped by the sun, Saved by the side: The role of peer and adversarial effects in adaptation to heat

PUBLICATIONS

BOOK CHAPTER

Chatterjee, D., Lal, B., & Saxena, R. (2021). [Education: Understanding the Gender Gap in Education and Employment](#). In S. Chakravorty & N. Sircar (Eds.), *Colossus: The Anatomy of Delhi* (South Asia in the Social Sciences, pp. 273-289). Cambridge: Cambridge University Press. [doi:10.1017/9781108935654.013](https://doi.org/10.1017/9781108935654.013)

OTHER WRITING

Arora, K., Saxena, R. (2018). [India-China Economic Relations: An Assessment](#). *South Asian Voices*, Stimson Center, Washington D.C

WORKS IN PROGRESS

1. *Climate Change and Mechanization of Agriculture in India* (with [Bhavya Srivastava](#))
 2. *Climate Change and Informal Insurance Networks* (with [Matthew Gordon](#))
 3. *Energy cost pass through in Indian Manufacturing firms* (with [Johannes Gálle](#))
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TEACHING EXPERIENCE

University of Colorado Boulder

Instructor of Record: Natural Resource Economics

Aug 2024 - Present

Created, organized and teaching the course to 50 undergraduate students

Instructor of Record: Principles of Microeconomics

Aug - Dec 2022

Created, organized and taught the course to 200 undergraduate students

Managed four teaching assistants who each taught 25-student recitations

Average student rating: 4.69/6 (Economics Avg: 4.2/6)

Teaching Assistant

2019 - 2024

Principles of Microeconomics (x5 semesters)

Principles of Macroeconomics (x4 semesters)

Environmental Economics (x2 semesters)

AWARDS

Heartland Environmental and Resource Economics Workshop Travel Support, 2023

TWEEDS workshop Travel Grant, 2023 (\$525)

Faculty Support Fellowship, 2023 (\$515)

Graduate School Domestic Travel Grant, 2023 (\$450)

Graduate Award for Public Policy Research, 2022 (\$2,000)

Reuben A. Zubrow Fellowship in Economics, 2021 (\$2,000)

Continuing Education Teaching Fellowship, University of Colorado Boulder, 2021 (\$2,000)

Continuing Education Teaching Fellowship, University of Colorado Boulder, 2020 (\$1,000)

Scholarship, Department of Economics, University of Colorado Boulder, 2019 (\$10,000)

SEMINAR PRESENTATIONS

2024: 25th Annual CU Environmental and Resource Economics Workshop (*Scheduled*), Vail CO; AERE Session at Southern Economic Association Conference, Washington D.C. (*Scheduled*)

2023: 18th Annual Conference on Economic Growth and Development, Indian Statistical Institute, Delhi; Heartland Environmental and Resource Economics Workshop; 2023 Berkeley/Sloan Summer School in Environmental and Energy Economics; 98th Western Economic Association Conference; AERE Summer Conference; 12th Annual Front Range Energy and Environmental Economics Camp

2022: 23rd Annual CU Environmental and Resource Economics Workshop; 11th Annual Front Range Energy and Environmental Economics Camp

2021: Environmental Defense Fund

2019: Centre for Policy Research

PROFESSIONAL SERVICE

Association of Environmental and Resource Economists (AERE)

Organizing Committee, AERE sessions at Southern Economic Conference 2024

University of Colorado Boulder

Organizer, Economics Graduate Student Seminar Aug 2021 - Present

Host, Graduate Student Panel at Graduate Student Recruitment Spring 2024

Mentor, Graduate student Mentor (x2) 2021 - 2023

PROFESSIONAL MEMBERSHIPS

Association of Environmental and Resource Economists (AERE), Western Economic Association International (WEAI)

OTHER WORK EXPERIENCE

University of Colorado Boulder

Research Assistant (Dr. Taylor Jaworski) May - July 2022

Digitized manufacturing plant opening locations in US during World War II
Created data visualizations, and spatial equilibrium model

Research Assistant (Dr. Francisca Antman) May - July 2020

Antman, F. M., & Cortes, K. E. (2023). The Long-Run Impacts of Mexican American School Desegregation. *Journal of Economic Literature*, 61(3), 888-905.

Environmental Defense Fund

Pre Doctoral Fellow June - Aug 2021

Under Office of the Chief Economist
Collaborated with Economics, Science, Policy teams in USA and India on research on Economic impacts of Environment, and Environmental justice questions

University of Pennsylvania

Research Associate, Centre for Advanced Study of India (CASI) 2018 - 2019

In collaboration with Centre for Policy Research (CPR), Delhi

Vaishanv, Milan (2021). [Gender, social change, and urbanisation in North India](#). *International Growth Centre Project #IND-19069*

Khosla, R., Sircar, N., & Bhardwaj, A. (2019). Energy demand transitions and climate mitigation in low-income urban households in India. *Environmental Research Letters*, 14(9), 095008.

Bhardwaj, A., Joshi, M., Khosla, R., & Dubash, N. K. (2019). More priorities, more problems? Decision-making with multiple energy, development and climate objectives. *Energy Research & Social Science*, 49, 143-157.

[Small City Dreaming Project](#), Centre for Policy Research

SKILLS

Proficient: R, Julia, Quarto Markdown, GIS Tools, $\text{\LaTeX}$, C, C#

Intermediate: Python, Stata, Matlab, SQL

Languages: English (Native), Hindi (Native), Urdu, Punjabi

CITIZENSHIP

F1 Visa, Indian Citizen

REFERENCES

Jonathan Edward Hughes (advisor)

Department of Economics

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Boulder, CO 80309

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Taylor Jaworski (committee member)

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Boulder, CO 80309

Email: taylor.jaworski@colorado.edu

Daniel Kaffine (committee member)

Department of Economics

256 UCB

Boulder, CO 80309

Email: daniel.kaffine@colorado.edu

Beia Spiller (committee member)

Transportation Program Director

Resources for the Future

Washington DC, 20036

Email: bspiller@rff.org

WORKING PAPERS ABSTRACTS

Barriers within Borders: Structural Transformation and Climate Change in India (Job Market Paper)

This paper examines how climate change is hindering structural transformation in India and the role of internal trade barriers in it. I combine local temperature effects on productivity, consumption, and labor shares with a static spatial equilibrium model to evaluate the potential for labor reallocation out of agriculture as an adaptation to climate change. I use a novel district-level panel dataset created by sourcing data from diverse sectors of Indian economy. Empirical estimates show that long-run high temperature has more pronounced negative impact on agricultural than manufacturing (supply-side effect). I also estimate a negative impact of temperature on household expenditure and as Engel's law would predict this increases food expenditure of households (demand-side effect). Further analysis shows that higher temperature increases agricultural labor share in India, inhibiting structural transformation. This phenomenon of increased specialization in low productivity agricultural sector in a developing country, like India, to maintain food security is known as "food problem". Districts in India could mitigate against the adverse effects of declining agricultural productivity through inter-district trade. Using a market access variable created by using development of Indian highways, I find that improved market access does not disrupt the positive relationship between temperature and agricultural labor share. This is primarily due to inefficiencies in Indian agricultural market due to a policy established in 1960s across states in India that restricts agricultural trade across state borders in India. I then calibrate a spatial equilibrium model with internal trade barriers to Indian data. Counterfactual analysis reveals that removing state-level trade barriers in Indian agriculture will increase income by 4.65% on average for each household and decrease agricultural labor share by 0.1pp on average for each district (~ 28.9 million people across India). This paper highlights the role of inefficient market policies in inhibiting adaptation to climate change, further indicating that internal political economy reforms may be as international trade policy in mitigating economic impacts of climate change.

Stumped by the sun, Saved by the side: Estimating the role of peer and adversarial effects in adaptation to heat

Literature has conclusively established that temperature has negative impact on individual's labor productivity. However we rarely work in isolation, most jobs require working with peers or against an adversary. This paper provides first estimates of the magnitude of peer and adversarial effect on individual's productivity under heat. Utilizing rich data, institutional details, and team dynamics of the sport of cricket, I find that even though temperature affects individual's productivity negatively, it doesn't have any effect on equilibrium outcomes that are affected by peers and adversaries. There could only be two explanations for this: increased peer effect under heat or a decreased adversarial effect. A further analysis reveals that peer effect increases significantly at temperature above 25C while adversarial effect has no significant difference between games played below 25C and above 25C temperature. These peer effects accrue through complementarity of skill-set among peers which creates opportunity for learning at higher temperature. This finding shows that even when workers are individually affected negatively by temperature, they can adapt in team settings through positive peer effect given complementary skills exist among peers.